Conservation of excitation energy to understand the dynamics of sun-induced fluorescence

David Martini et al.

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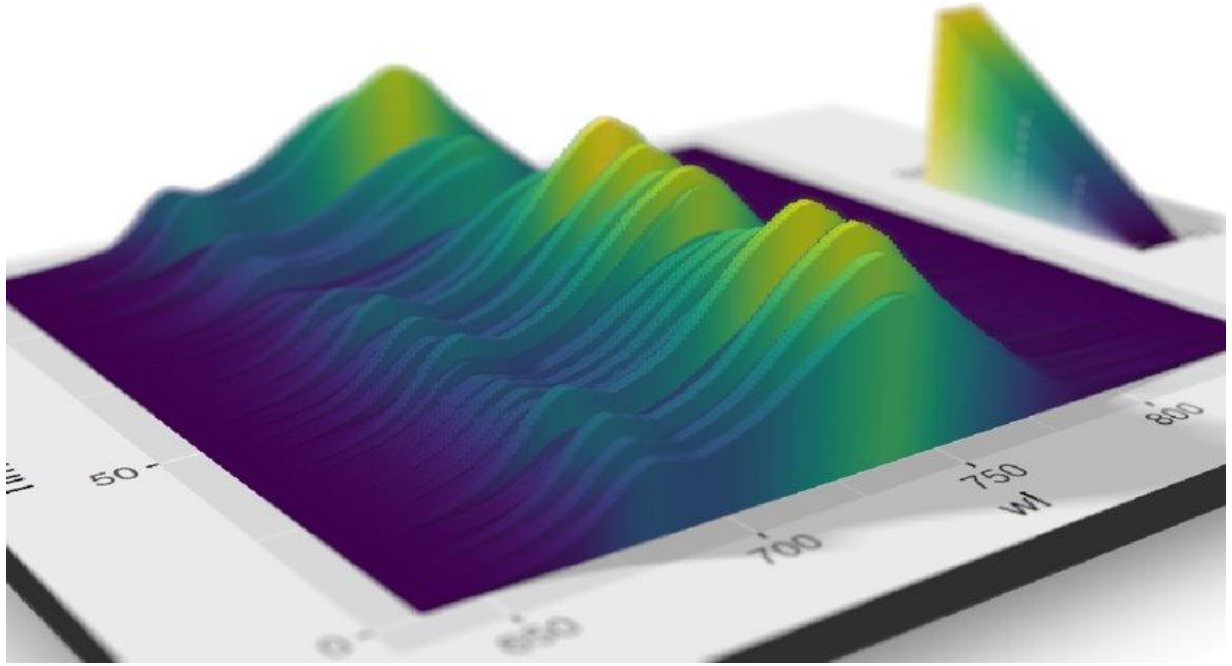
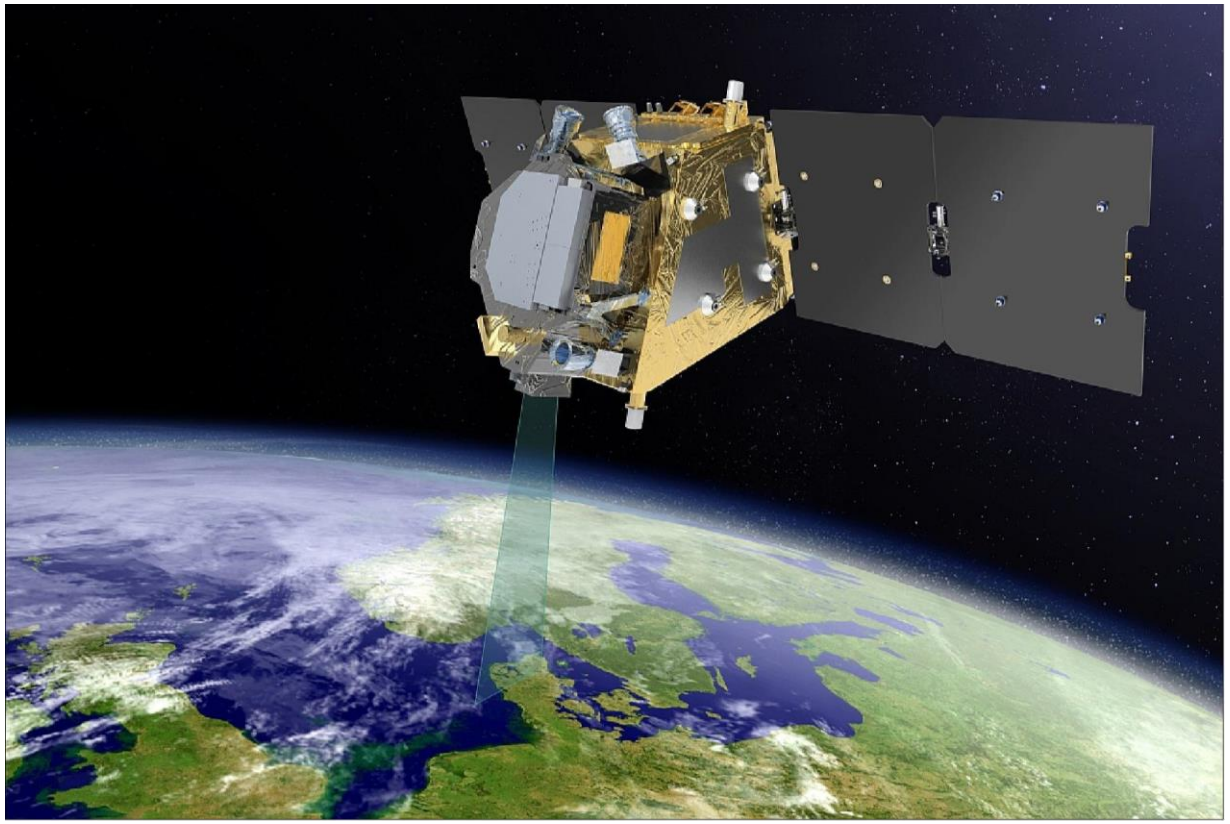
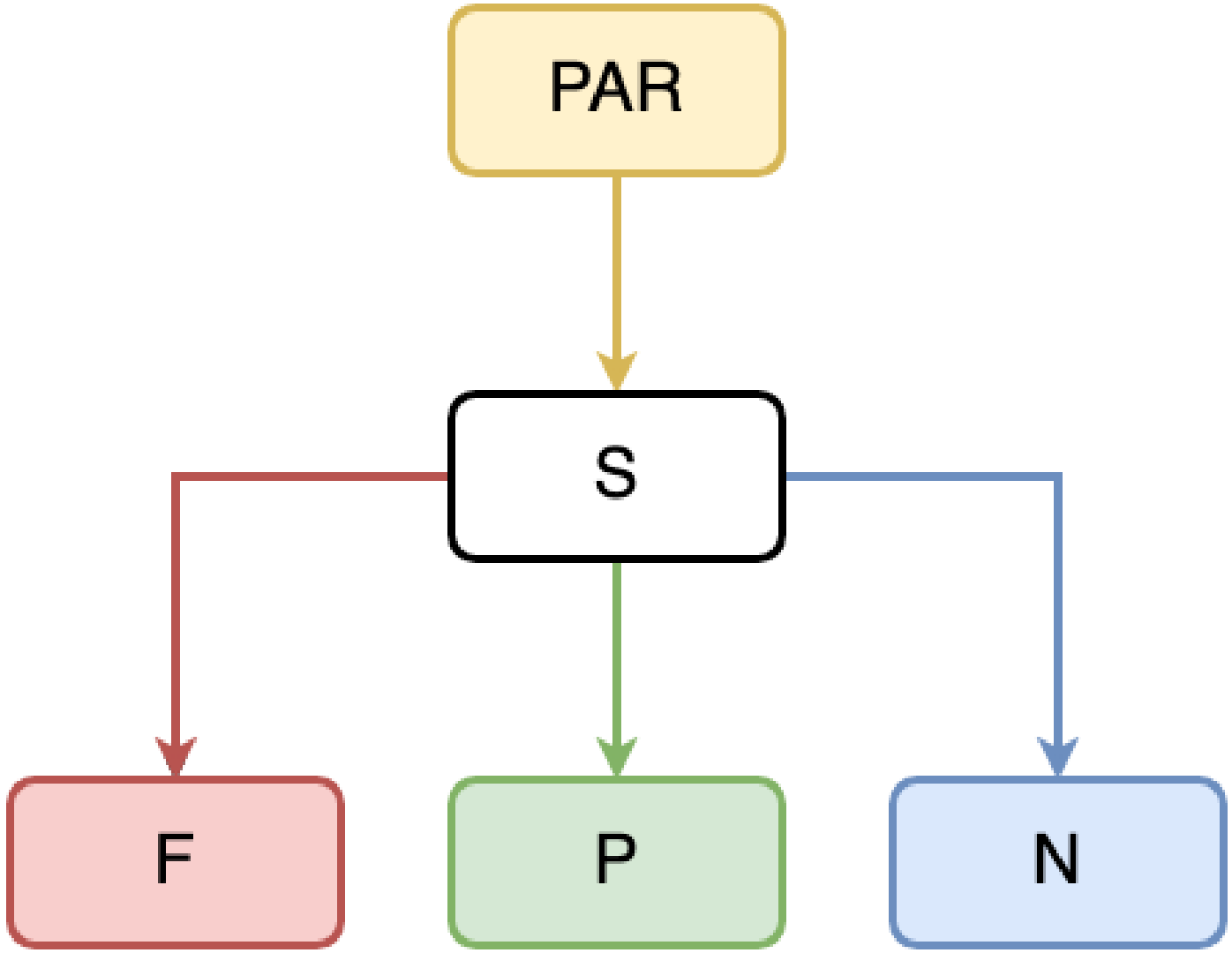
Sun-induced fluorescence (SIF)





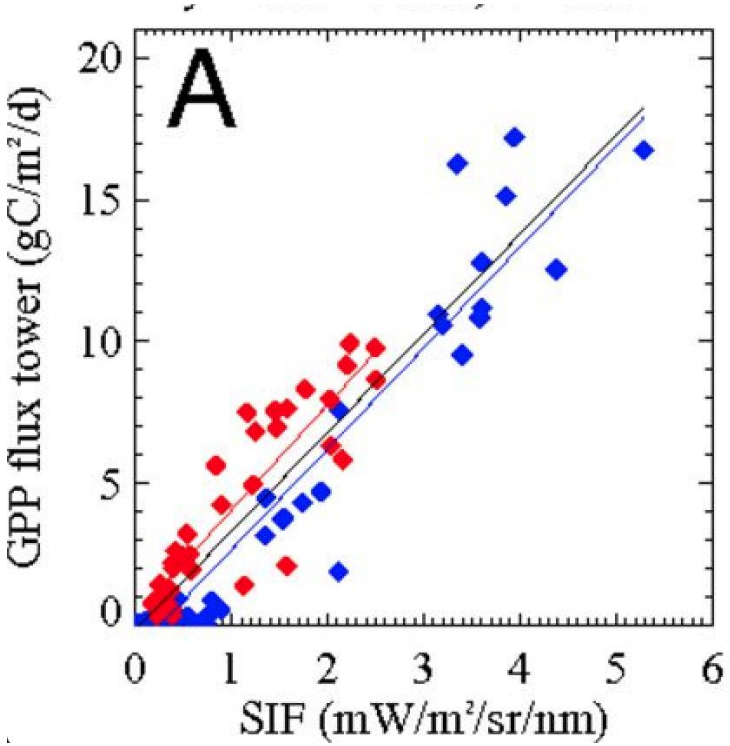
The promise of SIF

- To understand the allocation of excitation of energy and its dynamics

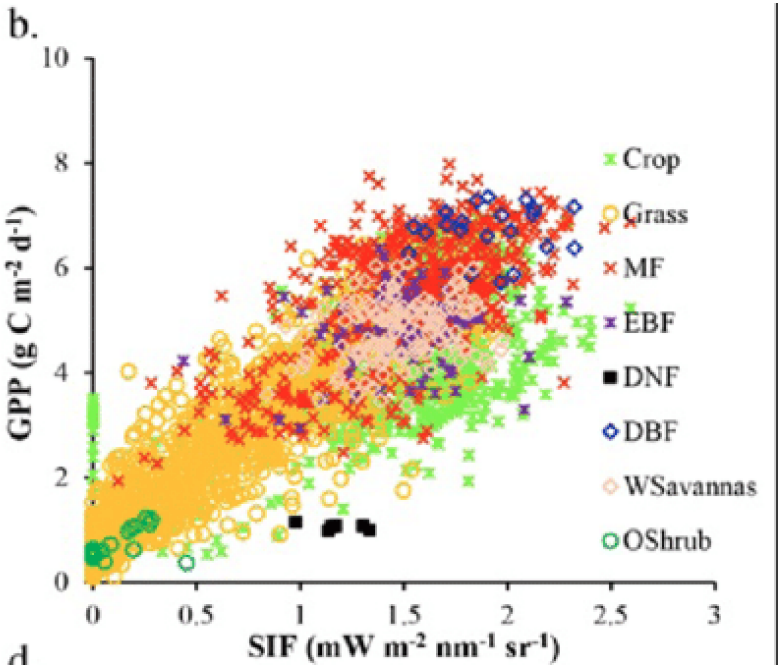


How SIF is mainly used

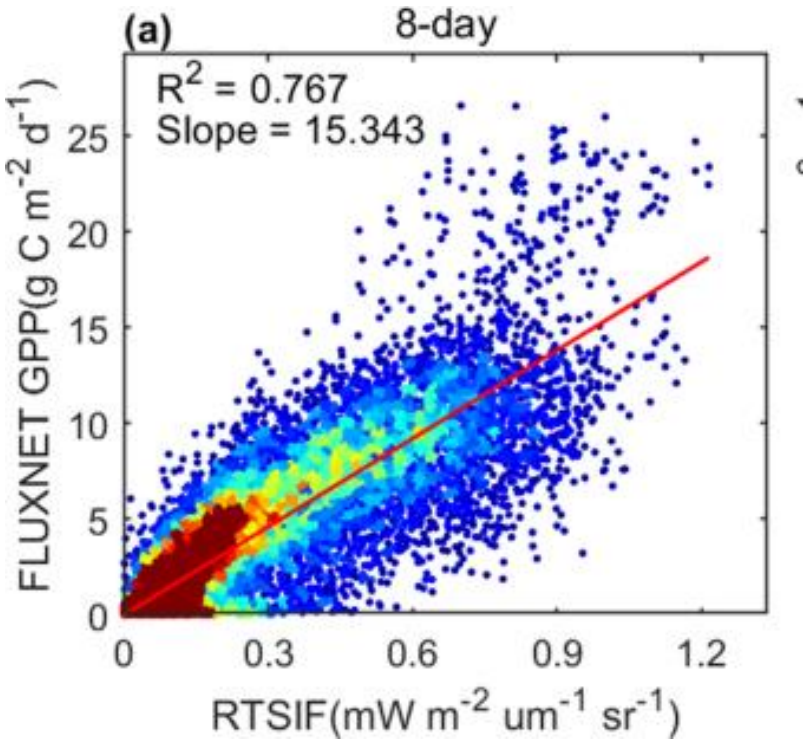
Guanter et al. 2014



Sun et al. 2017



Chen et al. 2022

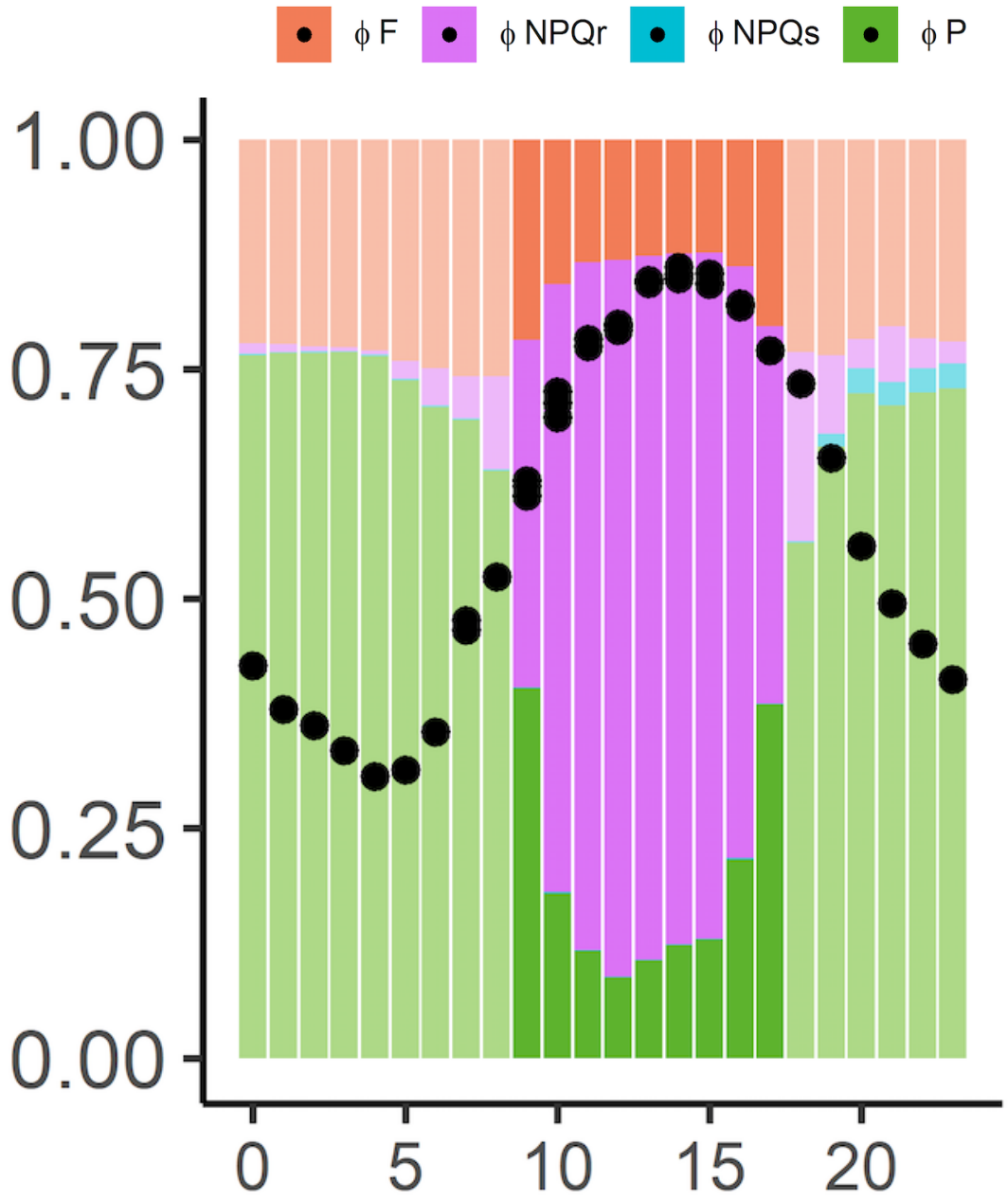


2 main challenges

1) The role of NPQ
2) Combining active and passive fluorescence

1) Role of NPQ

- NPQ can be the major dissipation pathway throughout the day



1) Role of NPQ

New Phytologist

Full paper | Open Access

Heatwave breaks down the linearity between sun-induced fluorescence and gross primary production

David Martini, Karolina Sakowska, Georg Wohlfahrt, Javier Pacheco-Labrador, Christiaan van der Tol, Albert Porcar-Castell, Troy S. Magney, Arnaud Carrara, Roberto Colombo ... See all authors

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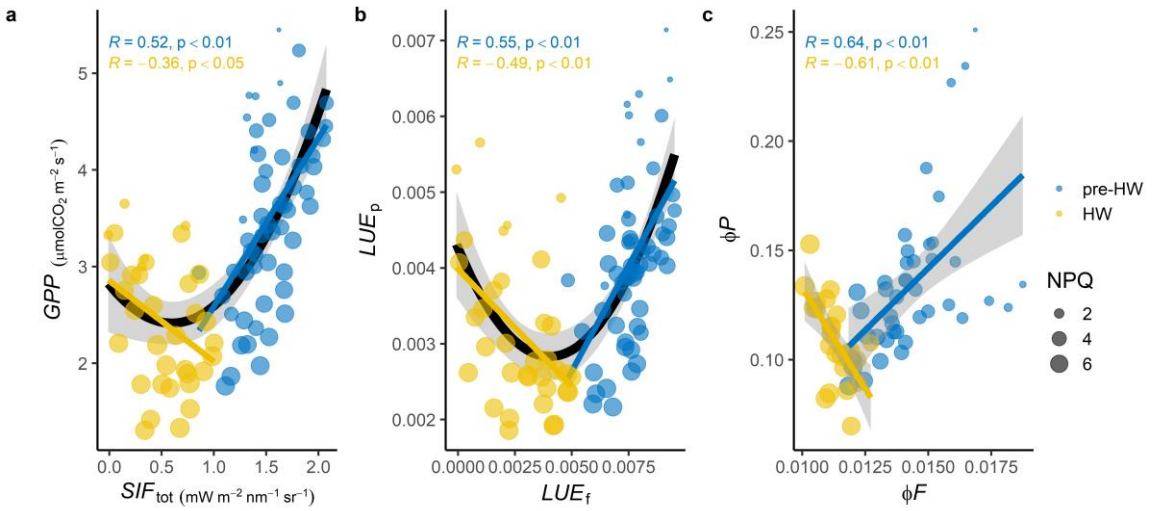
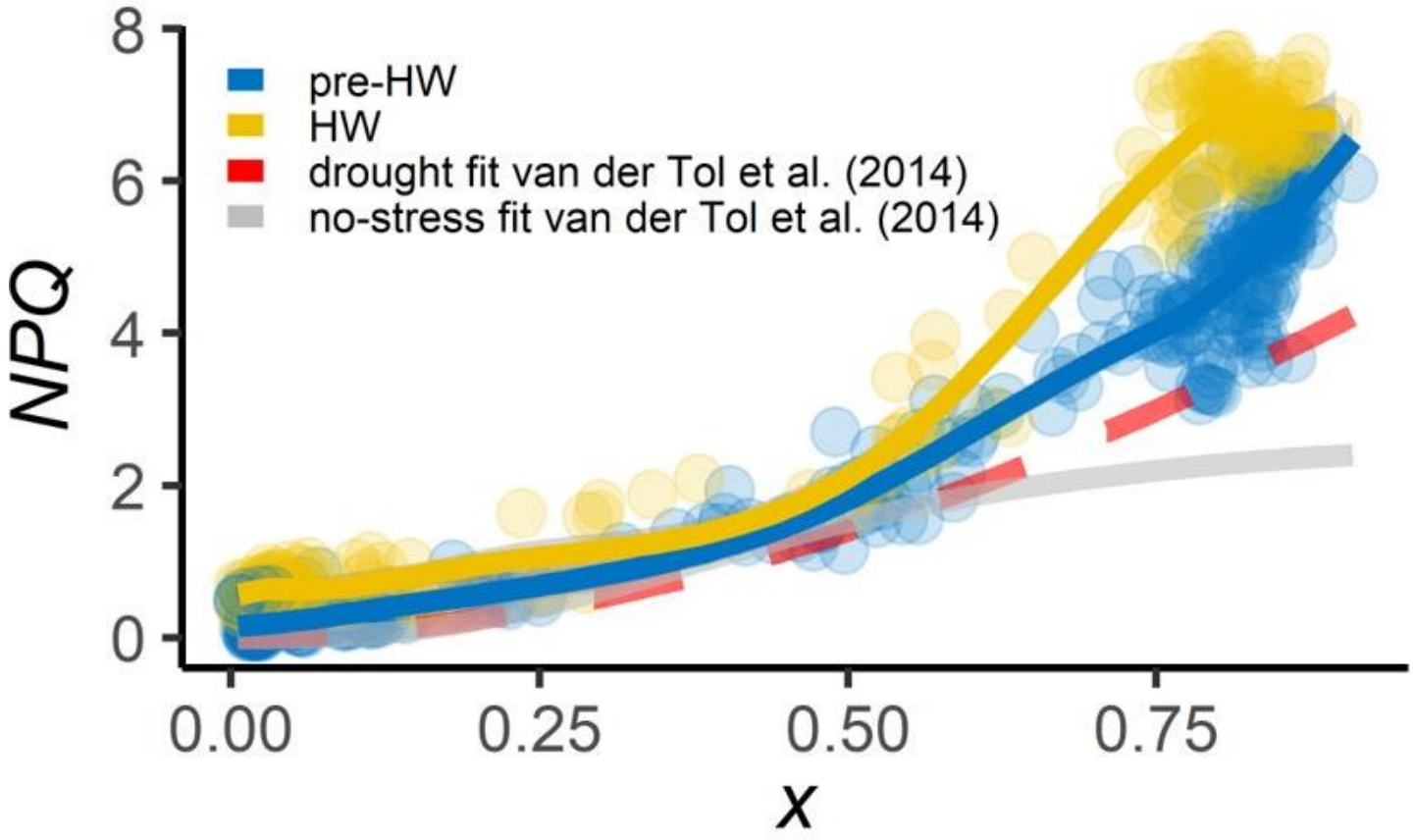


1) Role of NPQ

Comparison with SCOPE



ORCHIDEE TBM

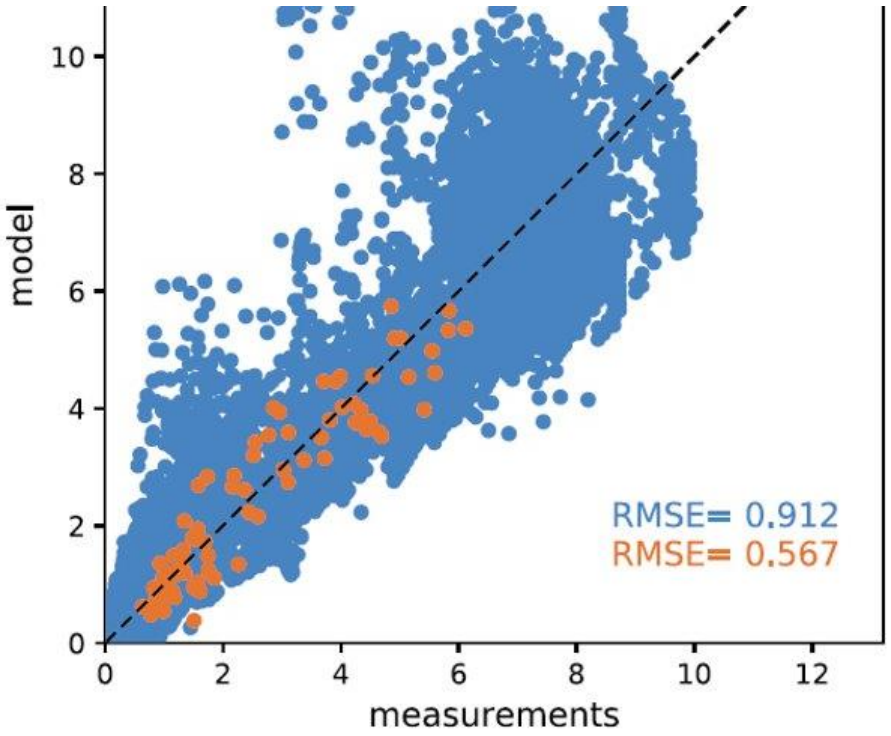
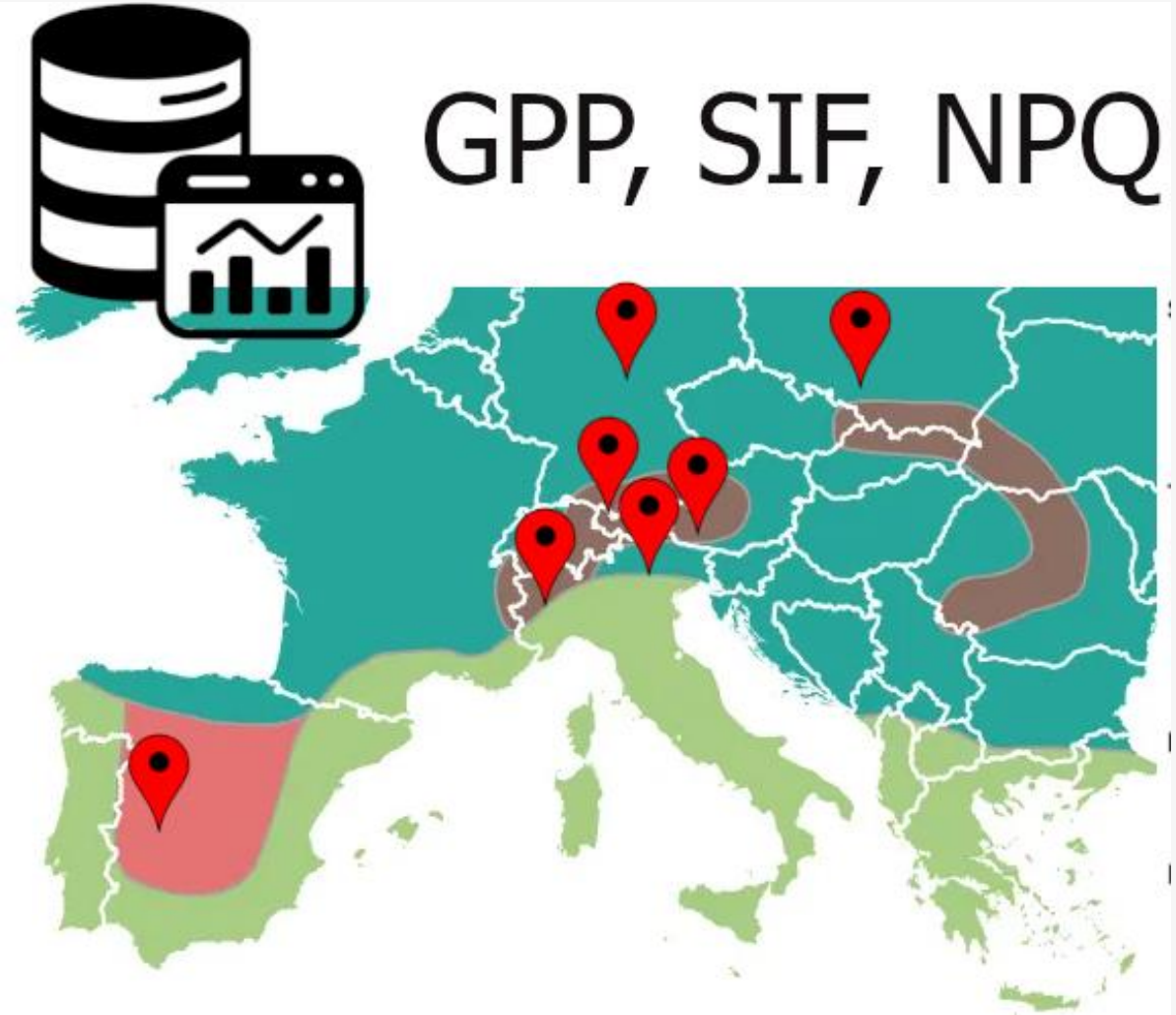


Figure 1. Modeled versus measured k_{NPQ} values (in s^{-1}) for the data sets of Porcar-Castell (2011)—in blue and Flexas et al. (2002)—in orange, with the corresponding RMSEs.

Bacour et al. 2019

2) Bringing together active and passive F

- Lack of simultaneous data



Austro-SIF project

- FAIR – Austria
- Plantadisch – Italy
- Torgnon – Italy
- Neustift- Austria
- Mezyk- Poland
- Majadas- Spain
- Leinefelde- Germany

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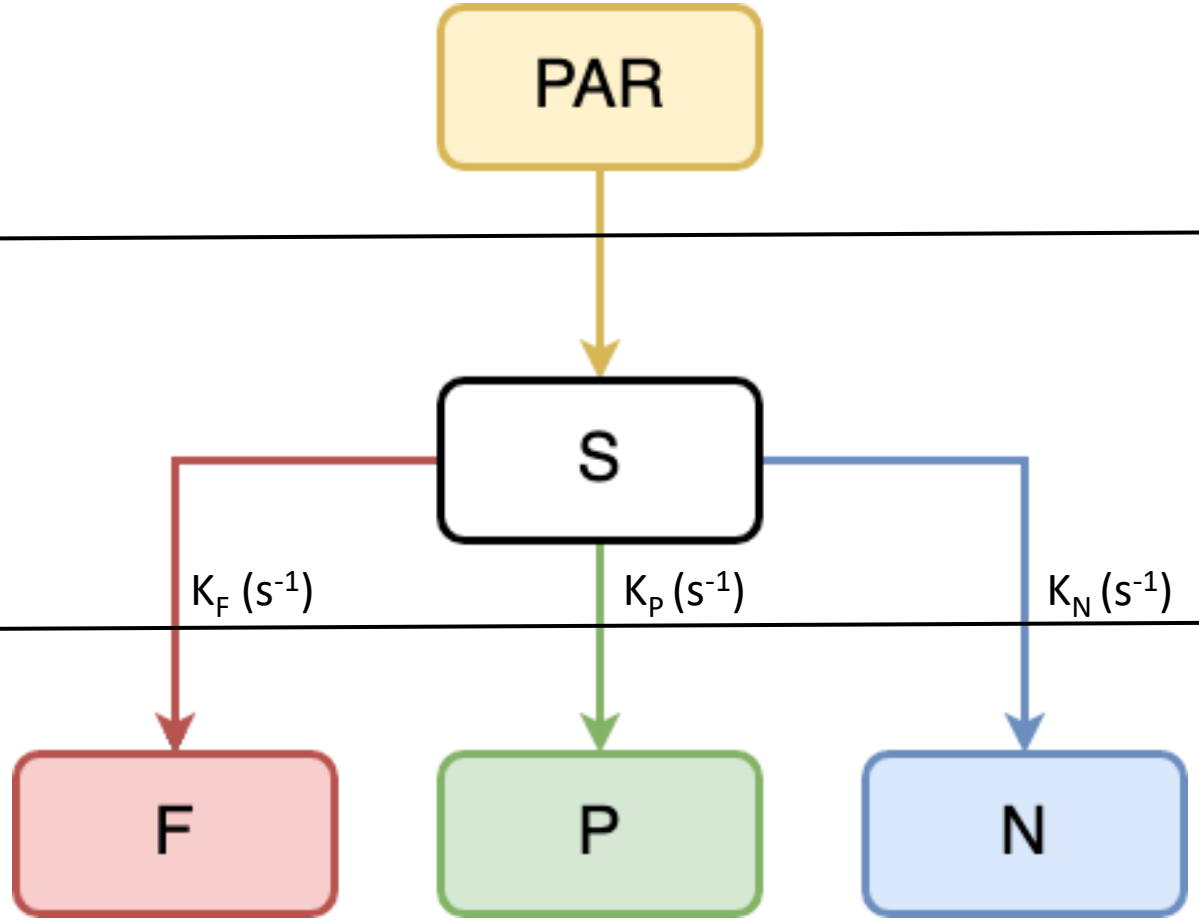
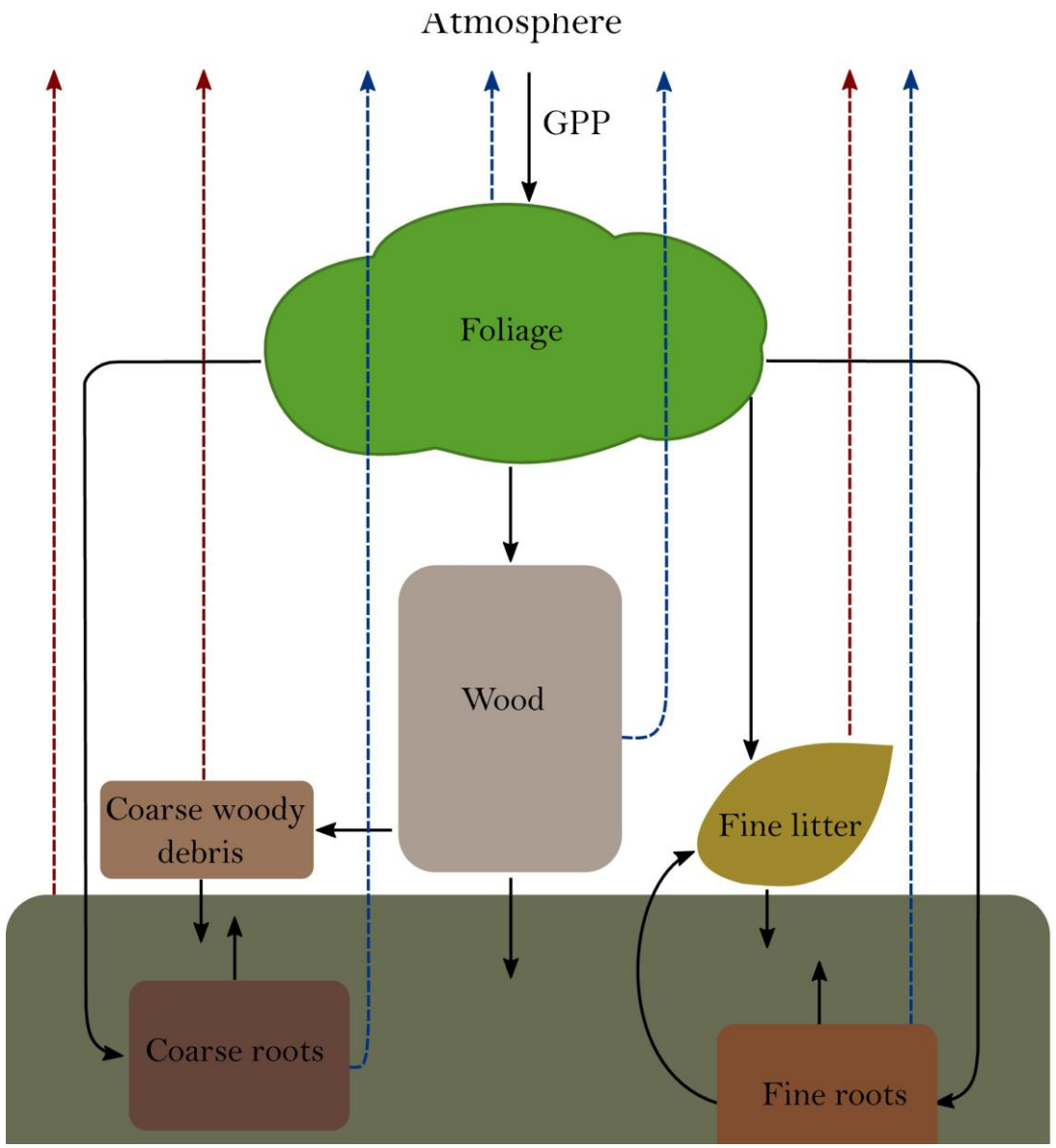
Allocation of excitation energy and its dynamics (preliminary)

- Incoming energy flux (mW m^{-2})
- Energy excitation (mJ m^{-2})
- Outgoing energy flux (mW m^{-2})



Similar approach used in other disciplines

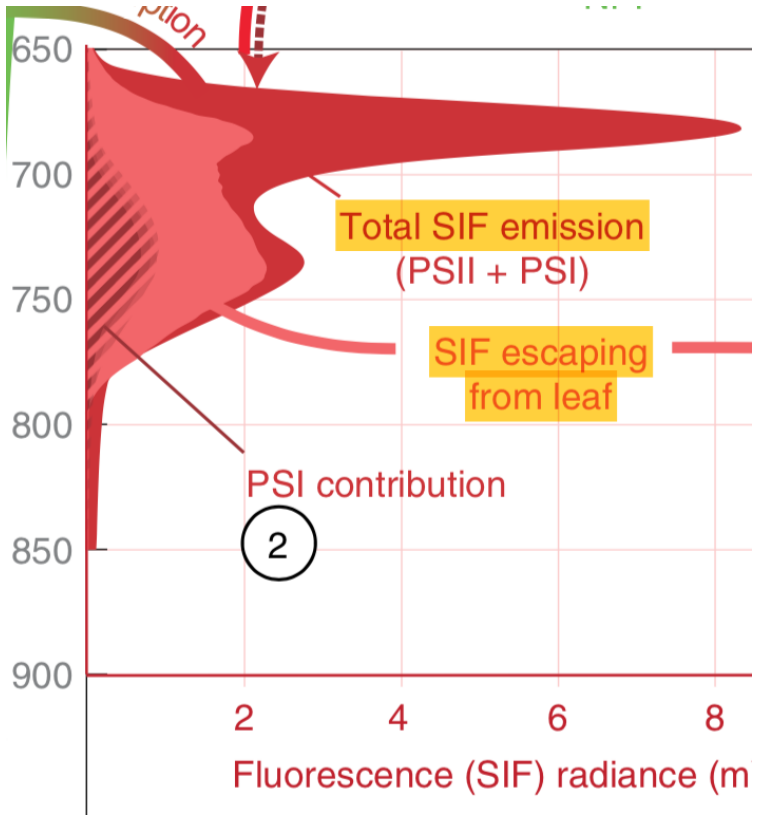
- S is assimilated carbon
- Allocated differentially (in time and space) to various compartments



Sierra et al. 2021

How to estimate F?

- F is SIF full spectrum at photosystem scale



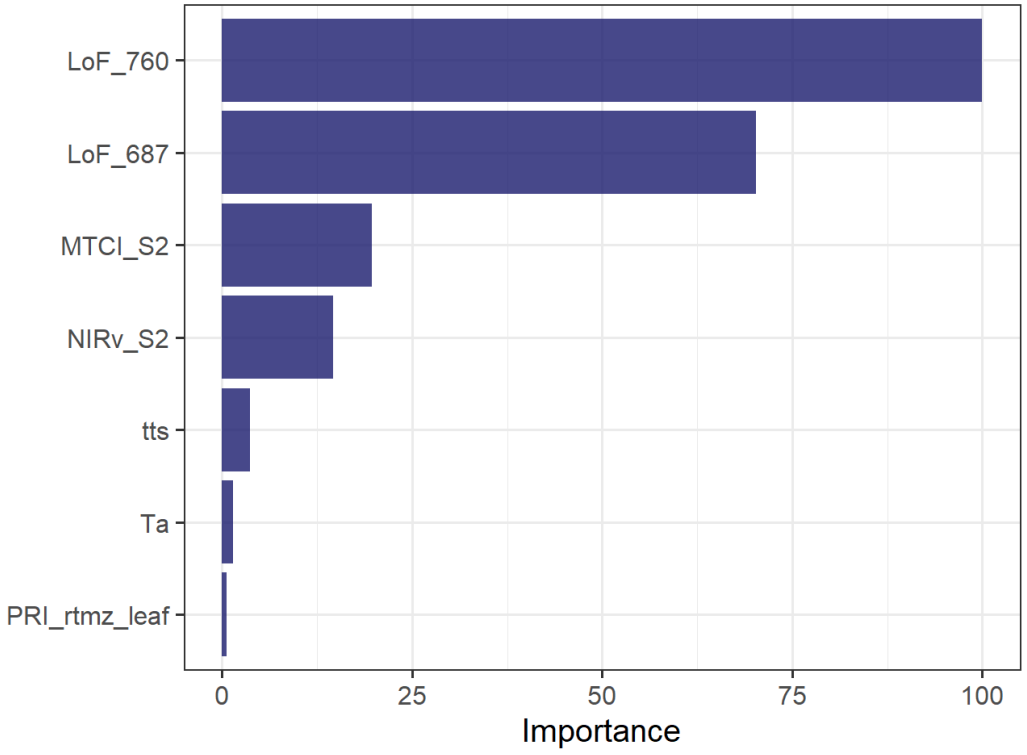
Porcar-Castell et al. 2021

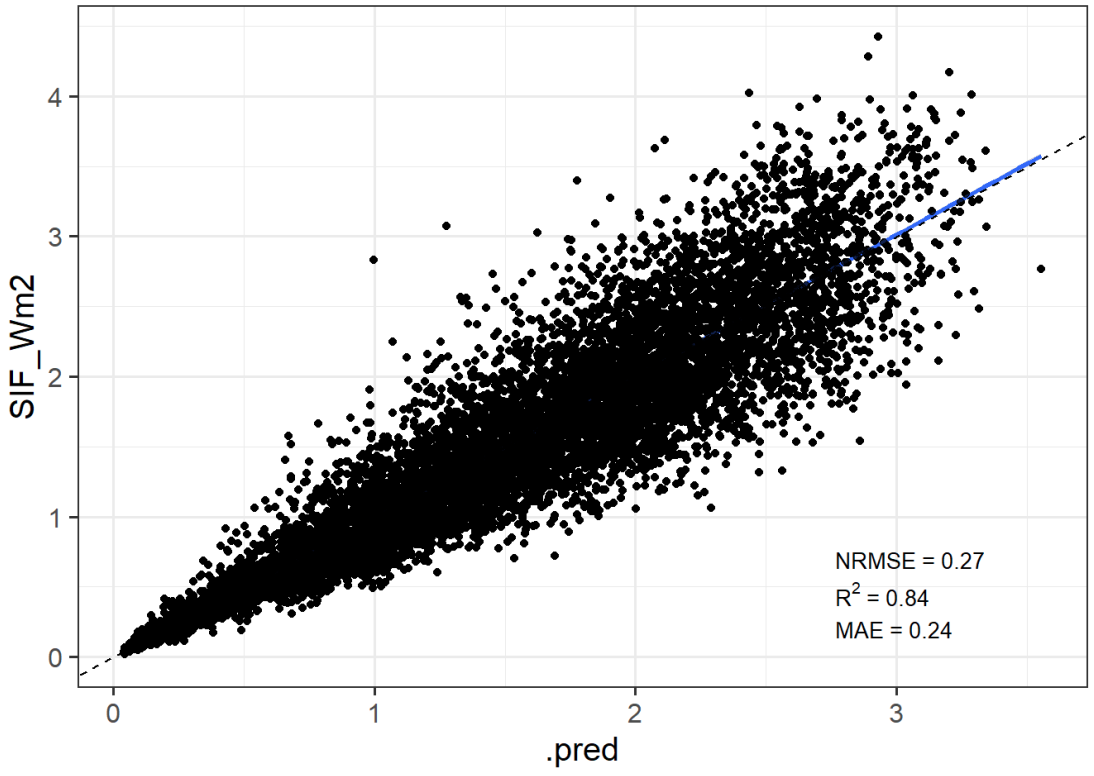
How to estimate F? Using SCOPE simulations

- Using SCOPE to generate a look-up table accounting for plan trait's covariance.
  - using data from spectral libraries and the TRY dataset to characterize the leaf traits covariance (pigments and the structural parameter) and trained a Gaussian Mixture Model to generate random samples of leaf traits that respected that covariance. Structural and soil parameters were drawn randomly (LHS).

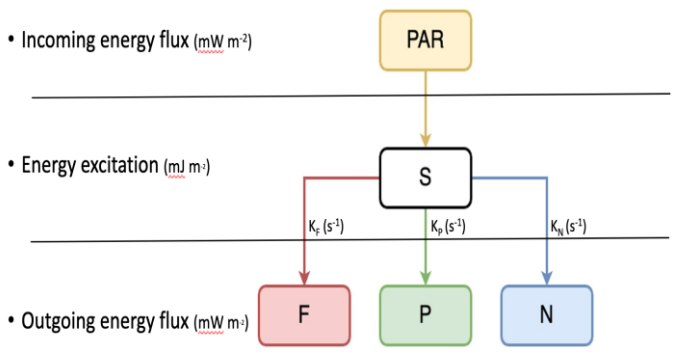
Javier Pacheco-Labrador

How to estimate F? SCOPE data + RF





The modelling approach (preliminary)



The relationship between the energy flux density of a dissipation process and its respective rate constant can be described as:

$$F = K_F \times S \tag{8}$$

$$P = K_P \times S \tag{9}$$

$$N = K_N \times S \tag{10}$$

$$D = K_D \times S \tag{11}$$

where K is a first order rate constant whose units are s^{-1} and S is the excitation energy whose units are $mJ\ m^{-2}$.

The variation of excitation energy (S) can be described as:

$$\frac{d(S)}{d(t)} = PAR \times \alpha - (K_F + K_P + K_N + K_D) \times S \tag{12}$$

Because these processes are very fast (occurring at ns scale) we can assume steady state ($\frac{d(S)}{d(t)} = 0$). Therefore all the incoming light energy is dissipated among the 4 competing processes.

Starting from the yields

dissipation (D) or by emitting fluorescence (F). Because these 4 processes compete for the same excitation energy the sum of their quantum yields (Φ) is equal to 1.

$$\Phi F + \Phi P + \Phi N + \Phi D = 1 \tag{5}$$

D is usually considered to be constant and part of N.

The quantum yield of a dissipation process can be defined as the first order rate constant (K) of that process divided by the total sum of the rate constants of the processes that can dissipate the excitation energy. For example for ΦF:

$$\Phi F = \frac{K_F}{(K_F + K_P + K_N + K_D)} \tag{6}$$

Alternatively, the quantum yield of a dissipation process can be defined by the ratio of the energy flux density of the process divided by the energy flux density of the photosynthetic active radiation absorbed by the antenna. For example for ΦF:

$$\Phi F = \frac{F}{(PAR \times \alpha)} \tag{7}$$

$$\frac{d(S)}{d(t)} = PAR \times \alpha - (K_F + K_P + K_N + K_D) \times S \tag{12}$$

By combining eq. 7 and eq. 5, you obtain eq. 12 at steady state

Using the actual rate constants

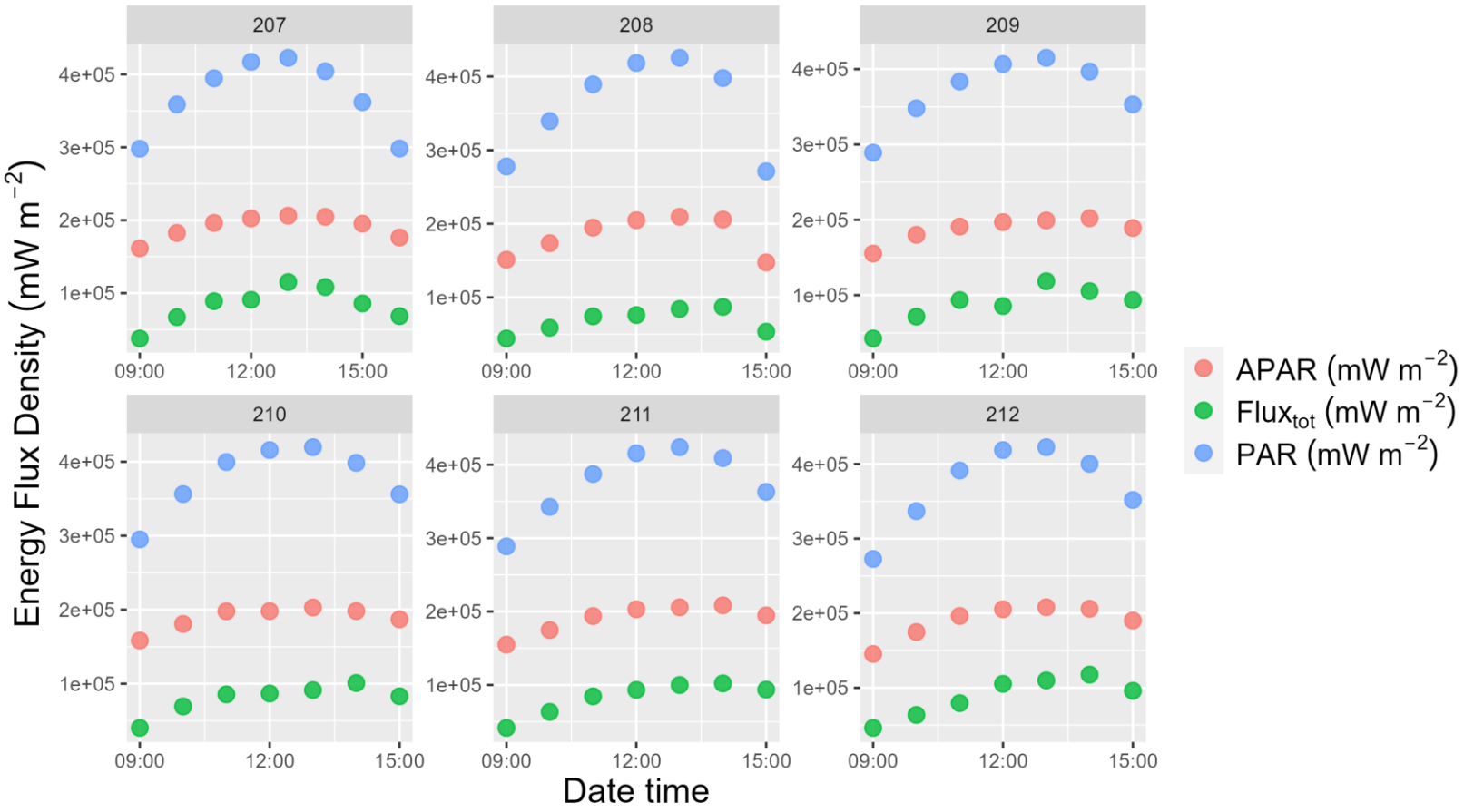
$$\frac{d(S)}{d(t)} = PAR \times \alpha - (K_F + K_P + K_N + K_D) \times S$$

$$\mathrm{NPQ} = \frac{k_{\mathrm{NPQ}}}{k_{\mathrm{D}} + k_{\mathrm{F}}}$$

Because K_D and K_F do not vary, in the active fluorescence literature it has generally assumed for convenience that $K_D + K_F = 1$.

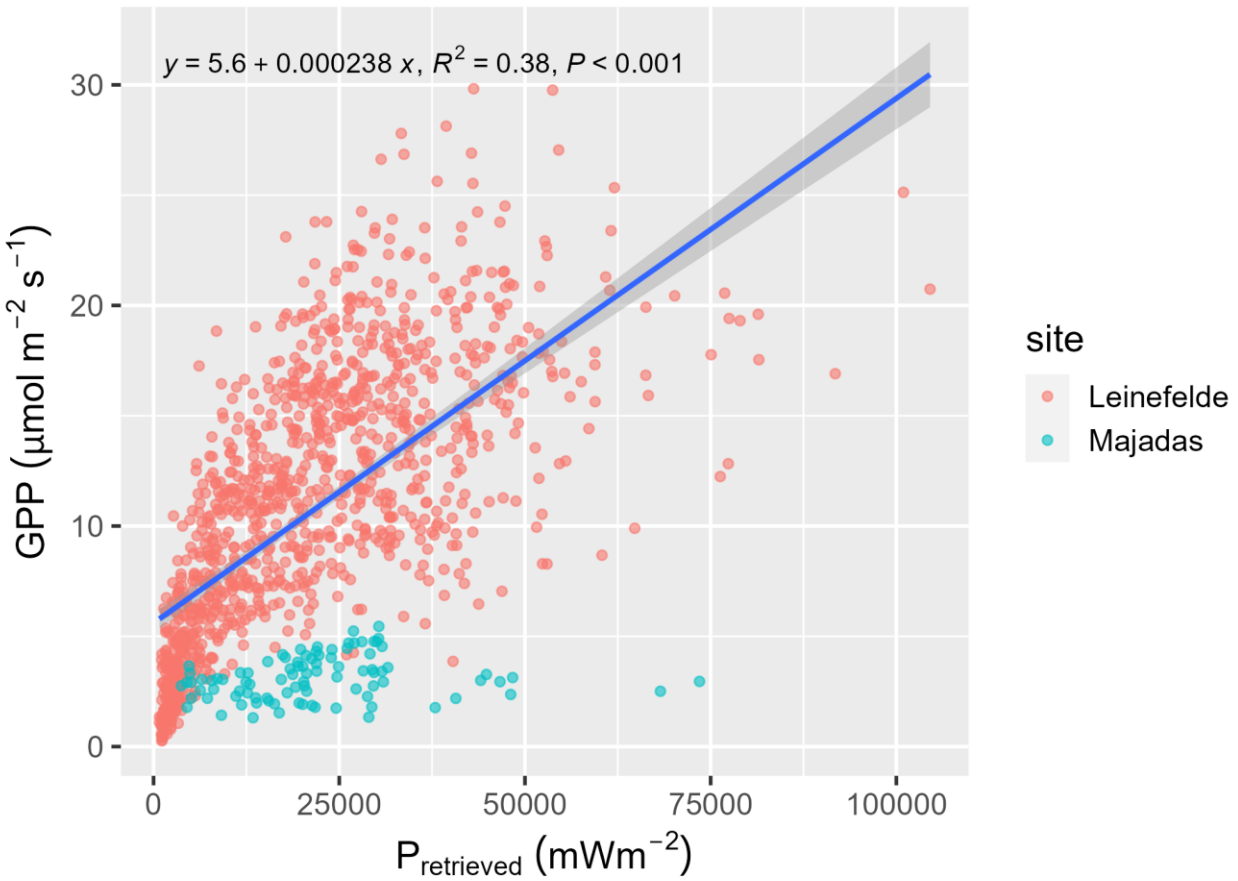
However in Lavergne and Trissl (1995) cf. Porcar-Castell *et al.* (2014) the values of these rate constants have been calculated by analyzing the fluorescence decay kinetics of isolated chloroplasts with laser pulses. The values reported are:

First results: energy budget in Majadas and Leinefelde



Fluxtot = F + P + N + D

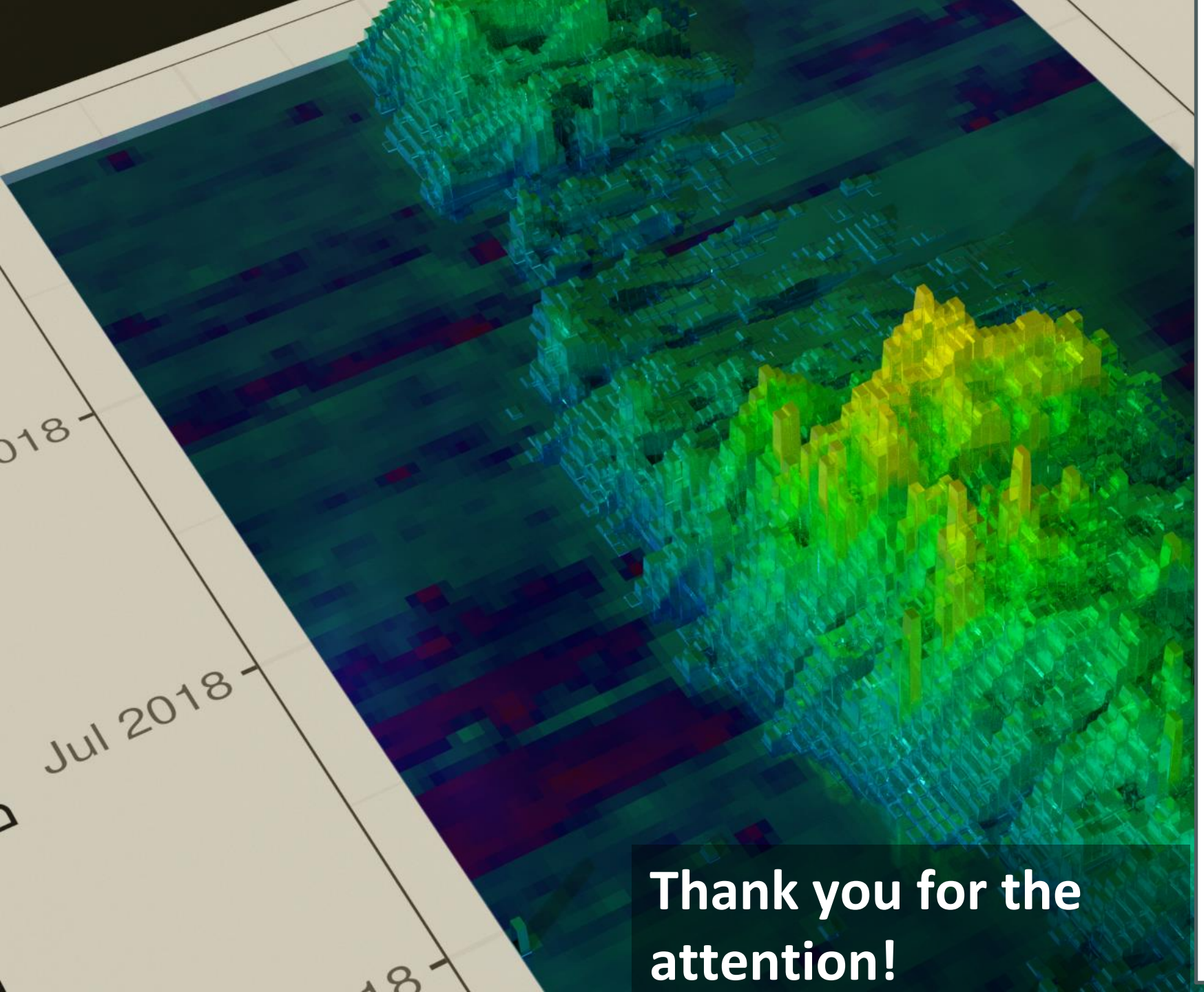
GPP-P



P only takes into account the light reactions of photosynthesis and it is therefore upstream of GPP. Any process that occurs in the dark reactions of photosynthesis, such as photorespiration, but also alternative energy sinks (such as circular electron transport, Meheler reaction or chlororespiration) can decouple the GPP-P relationship.

Uses and next steps

- If Kp and Kn are know the only unknown is alpha
- Alternatively both alpha and Kp can be obtained through an optimization procedure if Kn is known.
- Next steps:
  - Refine the model
  - Test with more sites (Austro-SIF dataset)



Thank you for the attention!

dmartini@bgc-jena.mpg.de